D07 · Order Management System (OMS/ATP/CTP)

05 · Order management analytics: fill rate, OTIF & service metrics

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Order management analytics measures how well orders are fulfilled — fill rate, OTIF, backorders, perfect order — by customer, SKU and channel, combining lagging results with leading risk indicators and translating them into revenue terms.

2 WHY IT MATTERS

Averages hide the problems that matter: a healthy overall fill rate can coexist with a handful of SKUs or key customers in serious trouble. **Segmented metrics show where to act; leading indicators show it in time; revenue translation gets the attention of leadership.** Perfect order rate is the most complete single view of what the customer experienced.

3 HOW IT IS DONE

1. **Define metrics consistently.** Fill rate, OTIF and perfect order with one formula and one data source each.
2. **Segment by customer, SKU and channel.** Find concentrations rather than reporting averages.
3. **Add leading indicators.** Orders at risk, aging backorders and late picks, visible daily.
4. **Translate gaps into money.** Revenue at risk and penalties, reviewed monthly with owners and actions.

IN PRACTICE

Report fill rate by SKU and customer — the global average hides problems

A 93% average fill rate can hide that 12 SKUs have a 72% fill rate and 3 key customers have 82% OTIF. Fill rate segmentation reveals actionable problems.

Translate fill rate and OTIF into financial language for the executive audience

Fill rate and OTIF are operational metrics. Lost revenue from stock-outs and OTIF penalties are financial metrics. The translation is what connects order management to management's business decisions.

Implement leading indicators (at-risk orders) in addition to lagging (what failed)

Lagging indicators (last month's fill rate) don't prevent the problem — they measure it afterward. Leading indicators (orders on backorder today with tomorrow's delivery date) allow action before the non-compliance occurs.

Automate the OMS dashboard — don't build it manually each month

A manually built Excel dashboard has a 5–10 business-day lag and error risk. A direct OMS connection via Power BI allows automatic, more frequent updates.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Order management dashboard — FMCG company, monthly management review — Monthly OMS dashboard for the Supply Chain Director's management review.

Order management KPI	Current value	Status vs. target
Fill rate % (total, all channels)	MEDIUM	93.4% vs. target 96% — improving from 91.8% previous month

Order management KPI	Current value	Status vs. target
OTIF % (strategic customers with SLA)	HIGH	88.2% vs. target 95% — 3 SLA non-compliances with accumulated penalties: $42K
Backorder rate % (orders on backorder)	MEDIUM	4.2% vs. target <2% — concentrated in 8 SKUs with insufficient safety stock
Perfect order rate % (complete + on-time + no damage + correct documentation)	HIGH	84.6% vs. target 92% — degraded by low OTIF

Result: Narrative: Fill rate improving but OTIF is low from 3 SLA non-compliances generating $42K in penalties. Root cause: 8 SKUs with insufficient safety stock generate 68% of backorders. Action plan: safety stock increase for 8 critical SKUs + review of priority dispatch process for SLA customers.

HOW IT IS MEASURED

Perfect Order Rate % (complete + on-time + no damage + correct documentation)

Perfect Order Rate %(Orders meeting all 4 criteria: completeness + OTIF + no damage + documentation / Total orders for the period) × 100

Benchmark: >92% perfect order rate in B2B distribution · >97% for premium SLA customers Perfect order rate is the most complete order management performance metric. A 97% fill rate with 91% OTIF and 1.5% damage rate results in a perfect order rate <89%.

Revenue at Risk from OTIF/Fill Rate Gap ($)

Revenue at Risk from OTIF/Fill Rate Gap ($)(Customer revenue × (1 - actual fill rate)) + (Customer revenue × (1 - actual OTIF) × % penalty)

Benchmark: $0 revenue at risk from OTIF/fill rate for strategic customers Quantifying revenue at risk converts fill rate and OTIF into financial metrics. “Fill rate is 2 pp below target” is an operational metric. “$240K revenue at risk” is a business metric.

WHAT YOU WOULD USE

OMS Analytics

Power BI — Order Management Dashboard — Module: Fill Rate + OTIF + Backorder AnalyticsPower BI dashboard connected to the OMS with order management KPIs: fill rate by SKU and customer, OTIF by channel and carrier, backorder aging, and perfect order rate.

SAP Analytics Cloud — Order Fulfillment Analytics — Module: SAP-Integrated Order Management KPIsSAP analytics platform with pre-configured fill rate, OTIF, and backorder reports integrated with SAP S/4HANA and SAP SD.

4 SO WHAT

Measure service where the customer feels it. One definition per metric, segmented views, leading indicators and a revenue translation that puts gaps on the leadership agenda.

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